Text embedding

universal-sentence-encoder-cmlm/en-base

Universal sentence encoder for English trained with conditional masked language model.

Publisher: Google License: Apache-2.0

Architecture: Transformer
Dataset: CommonCrawl Wikipedia
Language: English

Overall usage data
88.6k Downloads

Model formats

TF

TF2.0 Saved Model (v1)

Usage data: 88.6k Downloads

Fine tunable: Yes License: Apache-2.0 Format: TF2.0 Saved Model

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Download 388.11MB

Overview

The universal sentence encoder family of models map text into high dimensional vectors that capture sentence-level semantics. Our English-base (en-base) model is trained using a conditional masked language model described in [1]. The model is intended to be used for text classification, text clustering, semantic textural similarity, etc. It can also be use used as modularized input for multimodal tasks with text as a feature. The model can be fine-tuned for all of these tasks. The base model employs a 12 layer BERT transformer architecture.

Metrics

- We evaluate this model on SentEval sentence representation benchmark.

SentEval	MR	CR	SUBJ	MPQA	SST	TREC	MRPC	SICK-E	SICK-R	Avg
USE-CMLM-Base	83.6	89.9	96.2	89.3	88.5	91.0	69.7	82.3	83.4	86.0
USE-CMLM-Large	85.6	89.1	96.6	89.3	91.4	92.4	70.0	82.2	84.5	86.8

More details of the evaluations can be found in the paper [1].

To learn more about text embeddings, refer to the TensorFlow Embeddings documentation. Our encoder differs from word level embedding models in that we train on a number of natural language prediction tasks that require modeling the meaning of word sequences rather than just individual words. Details are available in the paper "Universal Sentence Representations Learning with Conditional Masked Language Model" [1].

Extended Uses and Limitations

The sentence embeddings can also be used for text classification, semantic similarity, clustering and other natural language tasks. Performance depends on the domain / data match of a particular downstream task to our pre-training data. Additional models are available from Universal Sentence Encoder family.

The model extends the BERT transformer architecture and it can be used for any tasks BERT can be applied. Performances gains over BERT depend on the particular task. The BERT model family contains additional BERT based models.

Example Use

Our model is based on the BERT transformer architecture, which generates a pooled_output of shape [batch_size, 768] with representations for the entire input sequences. The representation generated from this model is recommended to be used as is without fine-tuning. The model is also fine-tunable like other BERT models. It can be called like a regular BERT model as follows:

```
import tensorflow_hub as hub
import tensorflow as tf

english_sentences = tf.constant(["dog", "Puppies are nice.", "I enjoy taking long walks along the beach with my dog."])

preprocessor = hub.KerasLayer(
    "https://tfhub.dev/tensorflow/bert_en_uncased_preprocess/3")
encoder = hub.KerasLayer(
    "https://tfhub.dev/google/universal-sentence-encoder-cmlm/en-base/1")

english_embeds = encoder(preprocessor(english_sentences))["default"]

print (english_embeds)
```

References

[1] Ziyi Yang, Yinfei Yang, Daniel Cer, Jax Law, Eric Darve. Universal Sentence Representations Learning with Conditional Masked Language Model. EMNLP 2021